Instructions: Radio



Resource name	Radio
Description	This resource allows you to create a radio on which participants can tune to different frequencies and adjust the volume. The radio can be linked to an escape room puzzle and its solution.
Usage requirements	None.
Settings	
Skin	This setting allows you to select the appearance of the radio. It can take the following values: "Standard" or "Retro".
Action after solving	This setting allows you to select the action that will be carried out after the participants tune the radio to the correct station or frequency, that is, the station or frequency they need to tune the radio to in order to solve the linked puzzle. It can take the following values: **None** **Play sound**: If this action is selected, participants will be able to listen to an audio file on the radio after tuning to the correct station, and the linked puzzle will be completed once the audio has finished playing. This audio file must be specified through the “Stations” setting.
Loop station audio	This setting allows you to specify whether the audio files associated with the stations will play in a loop.
Initial frequency	This setting allows you to specify the initial frequency of the radio in MHz, with one decimal place. The value must be between 87.0 (or 88.0 for the “Retro” skin) and 108.0.
Initial volume	This setting allows specifying, in %, the initial volume level of the radio.

<table>
<tr><td>Stations</td><td>In this section, you can associate audio files with the stations that can be tuned in on the radio. For each station you want to associate with an audio file, you must specify the following:<table>
<tr><td>Frequency</td><td>It must be a number with one decimal place between 87.0 (or 88.0 for the “Retro” skin) and 108.0.</td></tr>
<tr><td>Audio</td><td>The URL of the audio file that will be associated with the station.
This audio file will play whenever the radio is tuned to the station's frequency or, if a tolerance greater than 0 has been specified in either the “Tolerance” or “Tolerance with noise” setting, whenever the radio is tuned to a frequency within the range around the station's frequency determined by these settings.
The audio file must be compatible with HTML5 and use one of the following formats: mp3, wav, ogg, oga, m4a, mp4, aac, webm, opus, or flac. In addition, its URL must end with the extension corresponding to the format used (e.g., .mp3, .wav, or .ogg).</td></tr>
</table></td></tr>
<tr><td>Tolerance</td><td>Maximum allowed frequency deviation for a station to be considered correctly tuned and for its associated audio to play without noise. For example, if this setting is set to 0.1, the audio file associated with a station with a frequency of 99.5 MHz will play without noise whenever the radio is tuned to a frequency between 99.4 and 99.6 MHz. Furthermore, if solving the linked puzzle requires tuning the radio to this station, participants will be able to solve the puzzle by tuning the radio to any frequency between 99.4 and 99.6 MHz.
The value of this setting must be a number with one decimal place. If its value is 0, the audio files associated with the stations will only play without noise when the radio is tuned to the exact frequency of the corresponding station, and, if the radio has a linked puzzle and there is no other tolerance greater than 0, solving it will require tuning the radio to the exact frequency specified by the solution.</td></tr>
<tr><td>Tolerance with noise</td><td>Maximum allowed frequency deviation for a station to be considered correctly tuned and for its associated audio to play with noise. Regardless of the value of this setting, whenever the radio is tuned to a frequency within the frequency range defined for a given station based on the value of the “Tolerance” setting, that station will be considered correctly tuned and its associated audio will play without noise. Similarly, regardless of the value of the “Tolerance” setting, whenever the radio is tuned to a frequency within the frequency range defined for a given station based on the value of the “Tolerance with noise” setting, that station will be considered correctly tuned and may therefore allow participants to solve the linked puzzle.
For example, if the “Tolerance with noise” setting is set to 0.4 and the “Tolerance” setting is set to 0.1, the audio file associated with a station with a frequency of 99.5 MHz will play without noise whenever the radio is tuned to a frequency between 99.4 and 99.6 MHz, whereas it will play with noise whenever the radio is tuned to a frequency between 99.1 and 99.3 MHz or between 99.7 and 99.9 MHz. Furthermore, if solving the linked puzzle requires tuning the radio to this station, participants will be able to solve the puzzle by tuning the radio to any frequency between 99.1 and 99.9 MHz.
The value of this setting must be a number with one decimal place.</td></tr>
<tr><td>Keep radio state</td><td>This setting allows preserving the radio state (the selected volume and, except for the “Retro” skin, the selected frequency) when the application is closed.
To avoid issues related to automatic audio playback in web browsers, the radio will always be turned off when the application starts, even if this setting is enabled. Since, with the “Retro” skin, the radio must be turned on using the volume dial, the volume will always be set to 0 when the application starts, even if this setting is enabled.</td></tr>
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Background	This setting allows you to change the default background. You can enter the URL of any image from your resource catalog.
Solution	You can link the radio to an escape room puzzle and specify a solution for that puzzle. If you do so, participants will be able to complete the puzzle by tuning the radio to a specific station or frequency. If solving the puzzle requires tuning the radio to a station, you can also require participants to listen to the audio file associated with that station until it finishes playing in order to complete the puzzle. See the description of the “Action after solving” setting for more details. The solution to the linked puzzle must be a frequency in MHz, expressed as a number with one decimal place. If this frequency matches the frequency of any of the stations specified in the “Stations” setting, participants will need to tune the radio to that station to solve the puzzle (which requires tuning the radio to the station's frequency or, if a tolerance greater than 0 has been specified in either the “Tolerance” or “Tolerance with noise” setting, to a frequency within the range around the station's frequency determined by these settings). Otherwise, participants will need to tune the radio to the exact frequency specified as the puzzle solution. For example, if participants need to tune the radio to a frequency of 96.1 MHz or to a station with that same frequency to solve the puzzle, the solution must be: *96.1*